

ECO 362: Financial Investments

Alpha

Motohiro Yogo

Princeton University

Types of registered investment companies

- Open-end mutual funds (\$22.1 trillion in 2022).
 - Number of shares adjust at the end of day, depending on buy and sell orders.
 - All orders are executed at NAV (closing price) at 4pm EST.
 - Holdings disclosed quarterly.
- Closed-end mutual funds (\$252 billion in 2022).
 - Fixed number of shares trade on exchanges.
 - Orders can be executed at any time during the trading day.
- Exchange-traded funds (\$6.5 trillion in 2022).
 - Fixed number of shares trade on exchanges.
 - Can short.
 - Holdings disclosed daily.

Market vs. limit order

- Open a mutual fund account to hold open-end funds.
- Need a brokerage account to trade closed-end funds and ETFs.
- Market order.
 - Buy up to Q shares at ask price from willing sellers.
 - Sell up to Q shares at bid price from willing buyers.
- Limit order.
 - Buy up to Q shares at price at most P .
 - Sell up to Q shares at price at least P .
- A limit order may not execute (at least immediately) if P is too low or high.

Types of mutual funds and ETFs

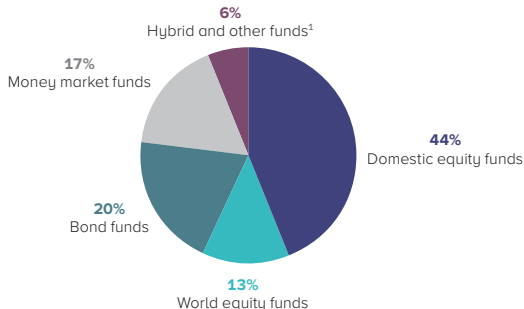
- Equity mutual funds.
 - Domestic vs. international.
 - Value vs. growth.
 - Large, mid, or small cap.
 - Sector.
- Bond mutual funds.
 - Government bonds.
 - Municipal bonds (tax-exempt).
 - Investment-grade bonds (rated BBB– and above by S&P or Baa and above by Moody's).
 - High-yield bonds.
- Money market funds.
 - Government bonds.
 - Municipal bonds (tax-exempt).

Composition of U.S. mutual funds and ETFs

FIGURE 2.2

The Majority of Investment Company Total Net Assets Were in Equity Funds

Percentage of total net assets, year-end 2022



Investment company total net assets:² \$28.9 trillion

¹ The *other funds* category includes ETFs—both registered and not registered under the Investment Company Act of 1940—that invest primarily in commodities, currencies, and futures.

² Closed-end fund data for total net assets include preferred share classes.

Actively managed vs. index funds

- Actively managed funds.
 - Investment mandate puts limits on types of securities that can be held. For example, a technology fund can only invest in technology stocks.
 - Within the mandate, portfolio choice is up to the investment manager.
 - A skilled manager can add value (but also charge higher fees).
- Index funds.
 - Tracks published indices such as S&P 500 or MSCI World Value Index.
 - Or passively managed following mechanical rules.
 - Could generate alpha relative to CAPM (e.g., value, low volatility, or momentum).
 - But nearly zero alpha relative to appropriate benchmark.

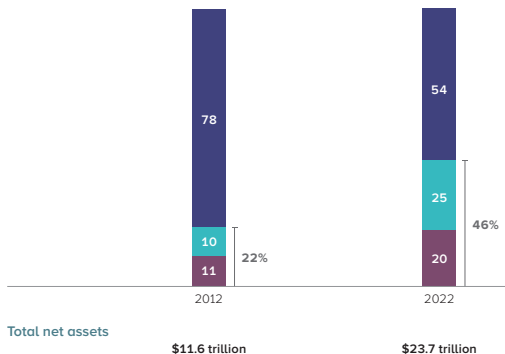
Growth of index funds and ETFs

FIGURE 2.5

Index Funds Have Grown as a Share of the Fund Market

Percentage of total net assets, year-end

- Actively managed mutual funds and ETFs
- Index ETFs
- Index mutual funds



Note: Data for ETFs exclude non-1940 Act ETFs. Data for mutual funds exclude money market funds.

Source: 2023 Investment Company Fact Book

Types of fees

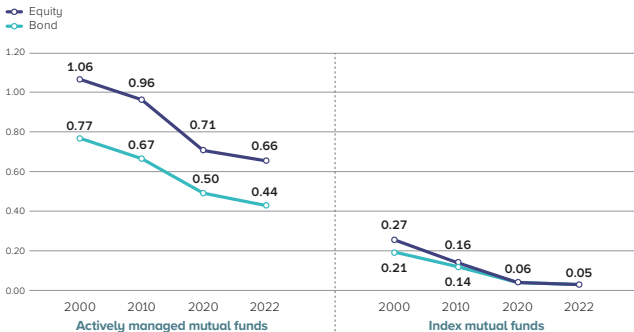
- Expense ratio: A percentage of account value is deducted to cover costs of portfolio management, administration, marketing, and distribution.
- $\text{Net return} = \text{Gross return} - \text{Expense ratio}$
- Loads.
 - Purchase fee is a percentage of account value at purchase.
 - Redemption fee is a percentage of account value at sale.
- Find no-load funds with no purchase and redemption fees (Vanguard and Fidelity).

Expense ratios of actively managed vs. index funds

FIGURE 6.6

Expense Ratios of Actively Managed and Index Mutual Funds Have Fallen

Percent



Note: Expense ratios are measured as asset-weighted averages. Data exclude mutual funds available as investment choices in variable annuities.

Sources: Investment Company Institute, Lipper, and Morningstar. See *ICI Research Perspective*, "Trends in the Expenses and Fees of Funds, 2022."

Source: 2023 Investment Company Fact Book

Distribution of mutual fund expense ratios

Type	Percentile			Average	
	10th	50th	90th	Asset-weighted	Simple
Equity	0.66	1.18	2.00	0.59	1.25
Index equity	0.06	0.33	1.53	0.09	0.61
Hybrid	0.65	1.15	1.98	0.70	1.26
Bond	0.45	0.81	1.61	0.48	0.93
Money market	0.17	0.40	0.66	0.25	0.40

Source: 2018 Investment Company Fact Book

- S&P 500 index funds: Expense ratios vary from 0.03% (Vanguard) to 1.62% (Rydex).
- Another failure of law of one price!

Zero-sum game

- The market overall earns zero alpha.
- Index funds or passive strategies must earn zero alpha relative to the appropriate benchmark.
- Some actively managed funds may earn positive alpha (before fees) at the expense of other actively managed funds.
- Or all actively managed funds may earn positive alpha (before fees) at the expense of retail investors or other institutions (pension funds, insurance companies, etc.)
- Even if gross alpha (before fees) is positive, net alpha (after fees) could be negative.

Performance of actively managed funds

Model	Alpha (% annual)		Beta			
	Net	Gross	Market	SMB	HML	WML
CAPM	-1.13 (-3.03)	-0.18 (-0.49)	0.99 (2.10)			
3-factor	-0.81 (-2.50)	0.13 (0.40)	0.96 (5.42)	0.07 (7.96)	-0.03 (-3.22)	
4-factor	-1.00 (-3.02)	-0.05 (-0.15)	0.97 (5.03)	0.07 (7.78)	-0.03 (-3.03)	0.02 (2.60)

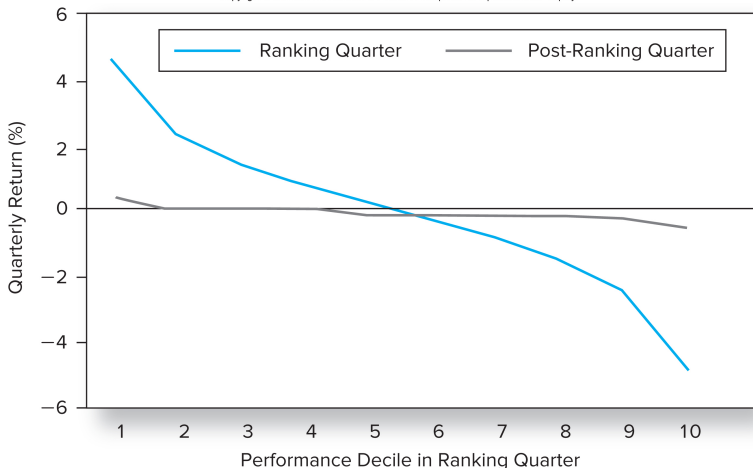
Source: Fama and French (2010, *Journal of Finance*)

Constructing portfolios sorted by prior returns

- Although actively managed funds collectively do not earn alpha, some funds may do so.
 - Skill or luck?
 - Skill should persist.
 - Evaluate funds relative to 3-factor model plus an additional WML (Winner Minus Loser) factor to account for momentum.
1. Sort mutual funds into 10 portfolios based on prior annual return (i.e., cutoff at each 10th percentile).
 2. Compute returns on these portfolios over the subsequent year.
 3. Rebalance annually.

Persistence of mutual fund performance

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Source: Bodie, Kane, and Marcus (2018, Figure 11.08)

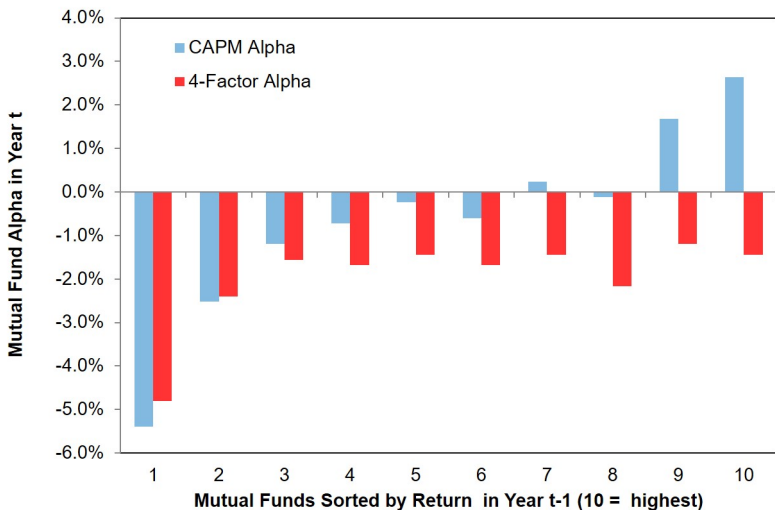
Note: x-axis is flipped so that 1 is the portfolio of mutual funds with the highest prior return.

Mutual funds sorted by prior returns

Statistic	Portfolio									
	Low	2	3	4	5	6	7	8	9	High
Mean (%)	0.01	0.23	0.34	0.36	0.40	0.38	0.45	0.43	0.59	0.68
SD (%)	4.90	4.60	4.48	4.30	4.36	4.35	4.41	4.56	4.72	5.04
Market beta	0.93	0.93	0.93	0.90	0.90	0.90	0.90	0.90	0.89	0.88
SMB beta	0.32	0.22	0.20	0.21	0.22	0.22	0.27	0.34	0.46	0.62
HML beta	-0.08	-0.10	-0.06	-0.03	-0.04	-0.05	-0.05	-0.07	-0.05	-0.05
WML beta	-0.09	-0.02	0.01	0.04	0.08	0.07	0.11	0.16	0.20	0.29
Alpha (%)	-0.40	-0.20	-0.13	-0.14	-0.12	-0.14	-0.12	-0.18	-0.10	-0.12
Alpha (<i>t</i> -stat)	-4.33	-3.11	-2.52	-3.09	-2.82	-3.31	-2.81	-3.65	-1.78	-1.60

Source: Cahart (1997, *Journal of Finance*)

Alpha on mutual funds sorted by prior returns



Source: Ang (2014, Figure 16.4)

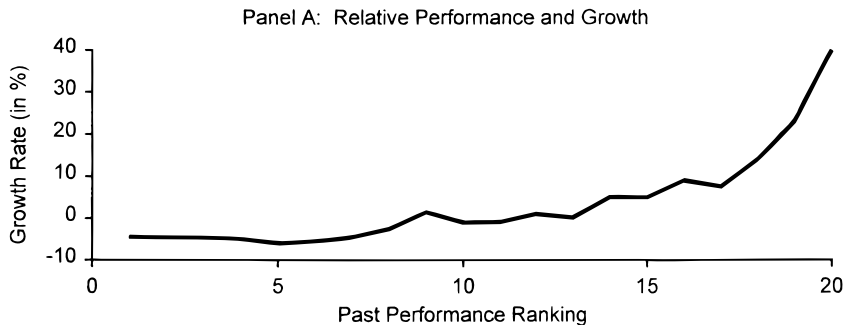
Explaining under-performance

- Persistence in performance relative to the CAPM but not the 4-factor model that includes the WML factor.
- Lucky funds that happen to hold winner stocks (i.e., high returns over the prior year) continue to do well because of momentum.
- Alpha is uniformly negative because of fees.
- Especially poor performance (i.e., funds in portfolio 1) is persistent.
- Cahart (1997) shows that poorly performing funds charge fees that are too high and trade too much.

Performance-flow sensitivity

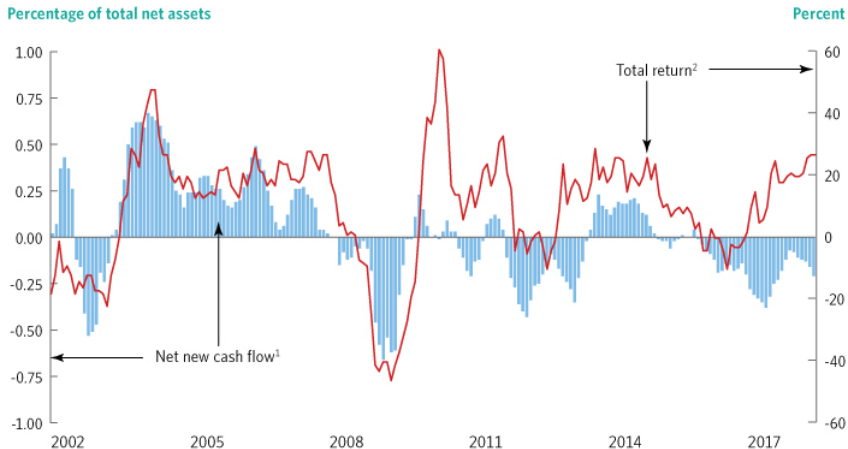
- Given that good performance does not persist, investors should not react to past performance when deciding which fund to invest in.
- However, investor flows react strongly to past performance in the cross section of mutual funds.
- In the aggregate, money flows into (out of) the mutual fund sector when returns are high (low).
- However, this is the opposite of rebalancing and also the market-timing strategy from the lecture on “Stocks”.
 - Optimal portfolio choice requires that investors go into mutual funds when the stock market falls.

Cross section of equity mutual fund flows



Source: Sirri and Tufano (1988, *Journal of Finance*)

Aggregate equity mutual fund flows



Source: 2018 Investment Company Fact Book

Does anyone have skill?

- Evidence presented thus far suggests that active management does not benefit investors.
- Any alpha that active management generates is more than offset by fees.
- But there are always exceptional performers in any industry.
 1. Fidelity Growth Company Fund.
 2. Berkshire Hathaway.
 3. Princeton University Investment Company (PRINCO).

Benchmark portfolio

- Grading on a curve: Compare scores across students taking the same course.
- Similarly, compare managers who have the same benchmark portfolio.
- Sometimes benchmarks are explicit.
 - S&P 500 index fund.
 - PRINCO's natural benchmark is Cambridge Associates' benchmark of university endowments.
- When not explicit, evaluate relative to 3-factor model.
 - Betas in 3-factor regression defines "what course?"
 - Alpha determines "what grade?"

Fidelity Growth Company Fund (FDGRX)

- Large-cap growth fund with expense ratio of 0.79%.
- Managed by Steven Wymer since 1997.
- Wymer was awarded by Morningstar as the 2017 U.S. Fund Manager of the Year.

Example 1: Evaluating the Fidelity Growth Company Fund

- 3-factor regression:

$$\begin{aligned} R_{i,t} - R_{b,t} = & \underbrace{0.29\%}_{t\text{-stat}=3.30} \\ & + 1.08(R_{m,t} - R_{b,t}) + 0.32 \times \text{SMB}_t - 0.45 \times \text{HML}_t \\ & + \epsilon_{i,t} \end{aligned}$$

- Long \$1.08 in market and short \$0.08 in riskless.
- $0.32 \times \text{SMB}$: Long \$0.32 in Small and short \$0.32 in Big.
- $-0.45 \times \text{HML}$: Short \$0.45 in High and long \$0.45 in Low.

Information ratio

- Breaking down performance.

$$\begin{aligned} R_{i,t} - R_{b,t} = & \underbrace{0.29\%}_{\alpha} \\ & + \underbrace{1.08(R_{m,t} - R_{b,t}) + 0.32 \times \text{SMB}_t - 0.45 \times \text{HML}_t}_{\text{benchmark}} \\ & + \underbrace{\epsilon_{i,t}}_{\text{tracking error}} \end{aligned}$$

- Skilled managers can generate alpha without deviating too much from the benchmark.
- Information ratio is

$$\frac{\alpha}{\sigma(\epsilon_{i,t})} = \frac{0.29\%}{1.88\%} = 0.15$$

- α : Average performance relative to benchmark.
- $\sigma(\epsilon_{i,t})$: Deviation from benchmark.

Berkshire Hathaway (BRK)



Warren Buffett (1930–)

CEO of Berkshire Hathaway and fifth wealthiest (\$118 billion in 3/2022).

You will notice that our major equity holdings are relatively few. We select such investments on a long-term basis, weighing the same factors as would be involved in the purchase of 100% of an operating business: (1) favorable long-term economic characteristics; (2) competent and honest management; (3) purchase price attractive when measured against the yardstick of value to a private owner; and (4) an industry with which we are familiar and whose long-term business characteristics we feel competent to judge.

Example 1 (continued): Evaluating Berkshire Hathaway

- 3-factor regression:

$$\begin{aligned} R_{i,t} - R_{b,t} = & \underbrace{0.88\%}_{t\text{-stat}=3.58} \\ & + 0.83(R_{m,t} - R_{b,t}) - 0.32 \times \text{SMB}_t + 0.44 \times \text{HML}_t \\ & + \epsilon_{i,t} \end{aligned}$$

- Long \$0.83 in market and long \$0.17 in riskless.
- $-0.32 \times \text{SMB}$: Short \$0.32 in Small and long \$0.32 in Big.
- $0.44 \times \text{HML}$: Long \$0.44 in High and short \$0.44 in Low.
- Information ratio is

$$\frac{\alpha}{\sigma(\epsilon_{i,t})} = \frac{0.88\%}{5.62\%} = 0.16$$

Trivia question

- Who is the most highly compensated employee of Princeton University?
- Hint: Total compensation was \$7.201 million in 2022.

Princeton University Investment Company (PRINCO)

- PRINCO manages \$37.7 billion (on June 30, 2021).
- Policy portfolio establishes guidelines for diversification across asset classes.

Asset class	Policy (%)	Actual (%)
Developed markets	13	7.6
Emerging markets	9	8.3
Hedge funds	24	21.1
Private equity	30	41.9
Real assets	18	13.8
Fixed income & cash	6	7.3

Example 2: Evaluating PRINCO

1. Policy portfolio.

$$R_{i,t} - R_{b,t} = \underbrace{2.16\%}_{t\text{-stat}=2.42} + 1.02(\text{Policy}_t - R_{b,t}) + \epsilon_{i,t}$$

- Same risk as policy portfolio, but added value both within and across asset classes.

2. Benchmark of 65% in S&P 500 and 35% in Barclays Government/Credit Bond Index.

$$R_{i,t} - R_{b,t} = \underbrace{5.45\%}_{t\text{-stat}=2.63} + 0.88(\text{Benchmark}_t - R_{b,t}) + \epsilon_{i,t}$$

- Lower risk and much higher return.

3. Cambridge Associates' benchmark of 160 university endowments.

$$R_{i,t} - R_{b,t} = \underbrace{4.16\%}_{t\text{-stat}=3.44} + 1.02(\text{CAB}_t - R_{b,t}) + \epsilon_{i,t}$$

- Similar risk and much higher return.

4. 3-factor model.

$$\begin{aligned} R_{i,t} - R_{b,t} = & \underbrace{5.75\%}_{t\text{-stat}=3.67} \\ & + 0.57(R_{m,t} - R_{b,t}) + 0.35 \times \text{SMB}_t + 0.00 \times \text{HML}_t \\ & + \epsilon_{i,t} \end{aligned}$$

- Low-beta small-cap strategy.

Summary of PRINCO's performance

- Added value from diversification into alternative asset classes such as hedge funds, private equity, and real assets.
- Ability to identify alpha within asset classes.
- Beats most other university endowments.
- What about “fees”?
 - In 2022, Andrew Golden (president of PRINCO) earned \$7.201 million.
 - Collectively, Golden and 5 managing directors earned \$20.431 million.
 - Total executive compensation of PRINCO as a share of the \$37.7 billion endowment is 0.05%.

Summary

- Shop around because fees vary across funds.
- Hard to find actively managed funds that outperform benchmarks, especially after fees.
- For most of us, better to stick to index funds with low fees.
- Don't chase past performance, and don't panic (and sell out) when the market falls.